E.g. If $V \subset \mathbb{CP}^n$ projective manifold then ω_{FS} restricted to V defines a Kähler metric.

A Kähler form ω on M defines deRham cohomology class $[\omega] \in H^2(M, \mathbb{R})$

<u>Theorem</u> ($\partial\bar\partial$-lemma) Let M be a compact Kähler manifold. Let ω, η be closed real $(1,1)$-forms such that $[\omega] = [\eta]$. Then $\exists f : M \to \mathbb{R}$ such that

$$\eta = \omega + \sqrt{-1}\,\partial\bar\partial f$$

<u>Theorem</u> (normal coordinates) If g is a Kähler metric then around any $p \in M$ $\exists$ holo. coordinates $z^1, \dots, z^n$ such that the components $g_{i\bar j}$ satisfy

$$g_{i\bar j}(p) = \delta_{ij} \quad , \quad \frac{\partial}{\partial z^k} g_{i\bar j}(p) = \frac{\partial}{\partial \bar z^k} g_{i\bar j}(p) = 0$$

Covariant derivatives and curvature

Let (M, g) be Kähler
ω associated 2-form (Kähler form)
∇ Levi-Civita connection so $\nabla g = 0$

In holo. normal coords J is constant so $\nabla J = 0$
$\omega(X,Y) = g(JX, Y)$ so $\nabla \omega = 0$.

In holo. coordinates $z^1, \dots z^n$ we let

$$\nabla_i = \nabla_{\frac{\partial}{\partial z^i}}, \quad \nabla_{\bar{i}} = \nabla_{\frac{\partial}{\partial \bar{z}^i}}, \quad \partial_i = \frac{\partial}{\partial z^i}, \quad \partial_{\bar{i}} = \overline{\partial_i} = \frac{\partial}{\partial \bar{z}_i}$$

<u>Note</u>: • $J(\nabla_j \partial_k) = \nabla_j J \partial_k = \sqrt{-1}\, \nabla_j \partial_k$

so $\nabla_j \partial_k \in T^{1,0}$.

Define Christoffel symbols Γ^i_{jk} by

$$\nabla_j \partial_k = \sum_{i=1}^n \Gamma^i_{jk} \partial_i$$

• $\nabla_{\bar{j}} \partial_k \in T^{1,0}$ and $\nabla_k \partial_{\bar{j}} \in T^{0,1}$

but ∇ is torsion free, so

$$\nabla_{\bar{j}} \partial_k - \nabla_k \partial_{\bar{j}} = [\partial_{\bar{j}}, \partial_k] = 0$$

$$\Longrightarrow \quad \nabla_{\bar{j}} \partial_k = \nabla_k \partial_{\bar{j}} = 0.$$

- $\nabla_{\bar{i}} T = \overline{\nabla_i \bar{T}}$ for any tensor

In sum ∇ is determined by the Γ^i_{jk}

and $\Gamma^{\bar{i}}_{\bar{j}\bar{k}} = \overline{\Gamma^i_{jk}}$. Torsion free : $\Gamma^i_{jk} = \Gamma^i_{kj}$

<u>E.g.</u> 1. $$\nabla_i dz^k = -\sum_{j=1}^{n} \Gamma^k_{ij} dz^j$$

2. Let us write a tensor such as

$$\sum_{j,\bar{k}} a_{j\bar{k}} dz^j \otimes d\bar{z}^k$$

as simply $a_{j\bar{k}}$.

Then

$$\nabla_{\bar{p}} a_{i\bar{j}} = \partial_{\bar{p}} a_{i\bar{j}} - \Gamma^{\bar{l}}_{\bar{p}\bar{j}} a_{i\bar{l}}$$ (sum over repeated indices)

Applied to metric tensor $g_{i\bar{j}}$, since $\nabla g = 0$

we have

$$\Gamma^{\bar{l}}_{\bar{p}\bar{j}} g_{i\bar{l}} = \partial_{\bar{p}} g_{i\bar{j}}$$

so using inverse $g^{i\bar{l}}$ we get

$$g^{i\bar{k}} \partial_{\bar{p}} g_{i\bar{j}} = \Gamma^{\bar{l}}_{\bar{p}\bar{j}} g_{i\bar{l}} g^{i\bar{k}} = \Gamma^{\bar{l}}_{\bar{p}\bar{j}} \delta^{\bar{k}}_{\bar{l}} = \Gamma^{\bar{k}}_{\bar{p}\bar{j}}$$

conjugate : $\Gamma^i_{jk} = g^{i\bar{l}} \partial_j g_{k\bar{l}}$

<u>Curvature</u> $R_i{}^j{}_{k\bar{l}}$ defined by

$$(\nabla_k \nabla_{\bar{l}} - \nabla_{\bar{l}} \nabla_k) \partial_i = R_i{}^j{}_{k\bar{l}} \partial_j$$

$$R_{i\bar{j}k\bar{l}} = g_{p\bar{j}} R_i{}^p{}_{k\bar{l}}$$

Note that $\nabla_k \nabla_l = \nabla_l \nabla_k$ and $\nabla_{\bar{k}} \nabla_{\bar{l}} = \nabla_{\bar{l}} \nabla_{\bar{k}}$

Ricci curvature is $R_{i\bar{j}} = g^{k\bar{l}} R_{i\bar{j}k\bar{l}}$

Scalar curvature: $R = g^{i\bar{j}} R_{i\bar{j}}$

<u>Some identities</u>

- $R_i{}^j{}_{k\bar{l}} = -\partial_{\bar{l}} \Gamma^j_{ki}$
- $R_{i\bar{j}k\bar{l}} = -\partial_k \partial_{\bar{l}} g_{i\bar{j}} + g^{p\bar{q}} (\partial_k g_{i\bar{q}})(\partial_{\bar{l}} g_{p\bar{j}})$

  $= -\partial_k \partial_{\bar{l}} g_{i\bar{j}}$ in normal coordinates

1st & 2nd Bianchi

- $R_{i\bar{j}k\bar{l}} = R_{i\bar{l}k\bar{j}} = R_{k\bar{j}i\bar{l}} = R_{k\bar{l}i\bar{j}}$
- $\nabla_p R_{i\bar{j}k\bar{l}} = \nabla_i R_{p\bar{j}k\bar{l}}$

- $(\nabla_k \nabla_{\bar{l}} - \nabla_{\bar{l}} \nabla_k)\, \partial_{\bar{i}} = - R^{\bar{J}}{}_{\bar{i} k \bar{l}}\, \partial_{\bar{J}}$

- $R_{i\bar{j}} = - \partial_i \partial_{\bar{j}} \log \det(g_{p\bar{q}})$

<u>Definition</u> Ricci form $Ric(\omega)$ is defined by $Ric(\omega) = \sqrt{-1}\, R_{i\bar{j}}\, dz^i \wedge d\bar{z}^j$

$Ric(\omega)$ is a closed real $(1,1)$-form

<u>Note</u>: if h is another Kähler metric then $\dfrac{\det(h_{p\bar{q}})}{\det(g_{p\bar{q}})}$ is independent of the choice of coordinates $\Longrightarrow \exists\, f : M \to \mathbb{R}$ such that $\dfrac{\det(h_{p\bar{q}})}{\det(g_{p\bar{q}})} = f$ in any chart.

Then
$$Ric(h) - Ric(g) = -\sqrt{-1}\, \partial\bar{\partial} \log f$$
so cohomology class $[Ric(g)]$ is independent of the metric.

<u>Definition</u> 1st Chern class $c_1(M) = \dfrac{1}{2\pi}[Ric(g)] \in H^2(M, \mathbb{R})$
$\in H^2(M, \mathbb{Z})$